Chapter 3. Homomorphisms and Factor Groups

Homomorphisms, factor groups, simple groups, group actions, Burnside counting

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3.1 Homomorphisms (§13)

Motivation. An isomorphism says two groups are the same; a homomorphism says one group is a "shadow" of another. For example, parity $S_n \to \{\pm 1\}$ forgets almost everything about a permutation but keeps its sign. The kernel measures exactly what is forgotten, and the results here show that the kernel controls the whole map.

A map $\phi : G \to G'$ of groups is a **homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all a, b. For $A \subset G$, $\phi[A]$ is its image; for $B \subset G'$, $\phi^{-1}[B]$ is its inverse image.

Example 3.1

The trivial homomorphism $g \mapsto e'$; $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$; the sign $S_n \to \{\pm 1\}$; reduction $\mathbb{Z} \to \mathbb{Z}_n$, $m \mapsto r$ (remainder); evaluation $f \mapsto f(c)$ on additive groups of functions; projections $G_1 \times G_2 \to G_i$; differentiation on C^1 functions.

Theorem 3.1 (13.12)

Let $\phi : G \to G'$ be a homomorphism.
(1) $\phi(e) = e'$.
(2) $\phi(a^{-1}) = \phi(a)^{-1}$.
(3) If $H \leq G$, then $\phi[H] \leq G'$.
(4) If $K' \leq G'$, then $\phi^{-1}[K'] \leq G$.
In short, ϕ preserves the identity, inverses and subgroups.

Proof. (1) $\phi(e)\phi(e) = \phi(ee) = \phi(e) = e'\phi(e)$; cancel.

(2) $\phi(a)\phi(a^{-1}) = \phi(e) = e'$.

(3) $\phi(a)\phi(b) = \phi(ab)$, $e' = \phi(e)$ and $\phi(a)^{-1} = \phi(a^{-1})$ lie in $\phi[H]$.

(4) If $\phi(a), \phi(b) \in K'$, then $\phi(ab) = \phi(a)\phi(b) \in K'$ and $\phi(a^{-1}) = \phi(a)^{-1} \in K'$; and $\phi(e) = e' \in K'$. ■

The **kernel** is $\ker \phi = \phi^{-1}[\{e'\}]$, a subgroup by (4).

Theorem 3.2 (13.15)

Let $H = \ker \phi$ and $a \in G$. Then $\phi^{-1}[\{\phi(a)\}] = aH = Ha$. In particular left and right cosets of H coincide.

Proof. $\phi(x) = \phi(a)$ iff $\phi(a^{-1}x) = e'$ iff $a^{-1}x \in H$ iff $x \in aH$. Likewise $\phi(x) = \phi(a)$ iff $\phi(xa^{-1}) = e'$ iff $x \in Ha$. ■

Corollary 3.3 (13.18)

A homomorphism is one-to-one iff its kernel is $\{e\}$.

Proof. If $\ker \phi = \{e\}$, the fibre of $\phi(a)$ is $a\{e\} = \{a\}$. Conversely, if ϕ is one-to-one, only e maps to $e' = \phi(e)$. ■

So to show ϕ is an isomorphism one checks the homomorphism property, $\ker \phi = \{e\}$, and surjectivity. A subgroup H is **normal**, $H \trianglelefteq G$, if $gH = Hg$ for all $g \in G$.

Corollary 3.4 (13.20)

$\ker \phi$ is a normal subgroup of G.

Proof. Theorem 3.2. ■

3.2 Factor groups (§14)

Motivation. Arithmetic modulo n makes $\mathbb{Z}$ into the smaller group $\mathbb{Z}_n$ by declaring multiples of n to be zero. Factor groups do the same for any group: collapse a subgroup H to the identity. This works exactly when H is normal, and the fundamental homomorphism theorem shows that every homomorphic image arises this way.

Theorem 3.5 (14.1)

Let $\phi : G \to G'$ be a homomorphism with kernel H. The cosets of H form a group G/H under $(aH)(bH) = (ab)H$, and $\mu : G/H \to \phi[G]$, $\mu(aH) = \phi(a)$, is an isomorphism. The product is well defined: it does not depend on the representatives a, b.

Proof. By Theorem 3.2, $aH = \phi^{-1}[\{\phi(a)\}]$, so μ is a well-defined bijection between the cosets and $\phi[G]$. Define the product of cosets by transporting the product of $\phi[G]$: the coset $\mu^{-1}(\phi(a)\phi(b)) = \mu^{-1}(\phi(ab)) = (ab)H$. Since this depends only on $\phi(a)$, $\phi(b)$, i.e. on the cosets, it is well defined, and G/H is a group isomorphic to $\phi[G]$ via μ. ■

Theorem 3.6 (14.4)

For $H \leq G$, left coset multiplication $(aH)(bH) = (ab)H$ is well defined iff $aH = Ha$ for all a.

Proof. Suppose it is well defined; let $x = ah \in aH$. Using the representatives x and a^{-1}, $(xH)(a^{-1}H) = (xa^{-1})H$, while also $(aH)(a^{-1}H) = eH = H$. So $aha^{-1} \in H$, i.e. $ah = h'a \in Ha$, and $aH \subset Ha$. Symmetrically $Ha \subset aH$ (use a^{-1} in place of a).

Conversely, if $aH = Ha$ for all a, and $a_1 = ah_1$, $b_1 = bh_2$, then $a_1b_1 = ah_1bh_2$; write $h_1b = bh_3$, so $a_1b_1 = ab(h_3h_2) \in (ab)H$. ■

Corollary 3.7 (14.5)

If $H \trianglelefteq G$, the cosets of H form a group G/H, the **factor group**, under $(aH)(bH) = (ab)H$.

Proof. The operation is well defined; associativity is inherited, $H = eH$ is the identity and $a^{-1}H$ is the inverse of aH. ■

Theorem 3.8 (14.9)

If $H \trianglelefteq G$, then $\gamma : G \to G/H$, $\gamma(x) = xH$, is a homomorphism with kernel H.

Proof. $\gamma(xy) = (xy)H = (xH)(yH)$, and $xH = H$ iff $x \in H$. ■

So the normal subgroups are exactly the kernels of homomorphisms.

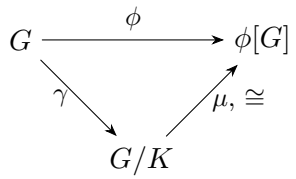


Figure. *The fundamental homomorphism theorem: $\phi = \mu \circ \gamma$, where $\gamma(g) = gK$ is the canonical map and $\mu(gK) = \phi(g)$ is an isomorphism.*

Theorem 3.9 (14.11, fundamental homomorphism theorem)

If $\phi : G \to G'$ is a homomorphism with kernel H, then $\phi[G]$ is a group and $\mu : G/H \to \phi[G]$, $\mu(gH) = \phi(g)$, is an isomorphism; with $\gamma(g) = gH$, $\phi = \mu \circ \gamma$.

Proof. Theorem 3.5, and $\mu(\gamma(g)) = \mu(gH) = \phi(g)$. ■

Example 3.2

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ via the reduction homomorphism. $\mathbb{R}/\mathbb{Z} \cong U$ via $x \mapsto e^{2\pi ix}$. $GL(n,\mathbb{R})/SL(n,\mathbb{R}) \cong \mathbb{R}^*$ via det.

Theorem 3.10 (14.13)

For $H \leq G$ the following are equivalent:
(1) $ghg^{-1} \in H$ for all $g \in G$, $h \in H$;
(2) $gHg^{-1} = H$ for all $g \in G$;
(3) $gH = Hg$ for all $g \in G$.

Proof. (1)⇒(2):

(1) gives $gHg^{-1} \subset H$; applying it to g^{-1}, $g^{-1}Hg \subset H$, i.e. $H \subset gHg^{-1}$.

(2)⇒(3): $gHg^{-1} = H$ gives $gH = Hg$ (multiply on the right by g).

(3)⇒(1): $gh \in gH = Hg$, so $gh = h'g$ and $ghg^{-1} = h' \in H$. ■

For fixed g, $i_g(x) = gxg^{-1}$ is an automorphism of G, the **inner automorphism** by g; condition (2) says H is invariant under all inner automorphisms.

3.3 Factor-group computations and simple groups (§15)

Motivation. Computing G/N is the main way to break a group into smaller pieces. Groups that cannot be broken further (no nontrivial proper normal subgroups) are the simple groups, the "primes" of group theory; here we see that A_n $(n \geq 5)$ is one of them.

Theorem 3.11 (15.8)

In $G = H \times K$, $\bar{H} = \{(h, e) : h \in H\}$ is normal and $G/\bar{H} \cong K$; similarly $G/\bar{K} \cong H$.

Proof. The projection $\pi_2(h,k) = k$ is an onto homomorphism with kernel $\bar{H}$; apply Theorem 3.9. ■

Theorem 3.12 (15.9)

A factor group of a cyclic group is cyclic.

Proof. If $G = \langle a \rangle$, then $(aN)^n = a^n N$, so $G/N = \langle aN \rangle$. ■

Example 3.3

$(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle (2,3) \rangle$ has order 12 and is generated by the images of $(1,0)$ (order 4) and $(0,1)$ (order 6, but $(0,3) \equiv (-2,0)$); it is isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_3 \cong \mathbb{Z}_{12}$. $(\mathbb{Z} \times \mathbb{Z})/\langle (1,1) \rangle \cong \mathbb{Z}$ via $(m,n) \mapsto m - n$. In S_3, A_3 is normal and $S_3/A_3 \cong \mathbb{Z}_2$.

A group is **simple** if it is nontrivial and has no proper nontrivial normal subgroups. Groups of prime order are simple.

Theorem 3.13 (15.15)

A_n is simple for $n \geq 5$.

Proof.

Step 1. A_n $(n \geq 3)$ is generated by 3-cycles: a product of two transpositions is a product of 3-cycles, since $(a,b)(c,d) = (a,c,b)(a,c,d)$ for distinct a,b,c,d and $(a,b)(a,c) = (a,c,b)$.

Step 2. If $N \trianglelefteq A_n$ contains a 3-cycle, then $N = A_n$: all 3-cycles are conjugate in A_n for $n \geq 5$. Indeed, given (a,b,c) and (a',b',c') choose $\sigma \in S_n$ with $\sigma(a) = a'$, $\sigma(b) = b'$, $\sigma(c) = c'$; then $\sigma(a,b,c)\sigma^{-1} = (a',b',c')$, and if σ is odd, replace it by $\sigma(d,e)$ with $d,e \notin \{a,b,c\}$, which gives the same conjugate.

Step 3. Let $N \neq \{\mathrm{id}\}$ be normal in A_n, $n \geq 5$; we find a 3-cycle in N. Let $\sigma \in N$, $\sigma \neq \mathrm{id}$, written as disjoint cycles.

Case 1: σ has a cycle of length $r \geq 4$, $\sigma = (a_1, \ldots, a_r)\mu$. With $\tau = (a_1, a_2, a_3)$, $N \ni \sigma^{-1}\tau\sigma\tau^{-1} = (a_1, a_3, a_r)$.

Case 2: σ has two 3-cycles, $\sigma = (a_1,a_2,a_3)(a_4,a_5,a_6)\mu$. With $\tau = (a_1,a_2,a_4)$, $\sigma^{-1}\tau\sigma\tau^{-1} = (a_1,a_4,a_2,a_6,a_3) \in N$, reducing to Case 1.

Case 3: $\sigma = (a_1,a_2,a_3)\mu$ with μ a product of disjoint transpositions; then $\sigma^2 = (a_1,a_3,a_2) \in N$.

Case 4: $\sigma = (a_1,a_2)(a_3,a_4)\mu$ with μ a product of disjoint transpositions. With $\tau = (a_1,a_2,a_3)$, $\alpha = \sigma^{-1}\tau\sigma\tau^{-1} = (a_1,a_3)(a_2,a_4) \in N$. Choose $i \notin \{a_1,\ldots,a_4\}$ (as $n \geq 5$) and $\beta = (a_1,a_3,i)$; then $\beta\alpha\beta^{-1} = (a_3,i)(a_2,a_4)$ and $\beta\alpha\beta^{-1}\alpha = (a_3,i)(a_1,a_3) = (a_1,i,a_3) \in N$.

In each case N contains a 3-cycle (the products are computed by following each symbol), so $N = A_n$ by Step 2. ■

Theorem 3.14 (15.16)

Let $\phi : G \to G'$ be a homomorphism. If $N \trianglelefteq G$, then $\phi[N] \trianglelefteq \phi[G]$. If $N' \trianglelefteq \phi[G]$ (or $N' \trianglelefteq G'$), then $\phi^{-1}[N'] \trianglelefteq G$.

Proof. For $\phi(g) \in \phi[G]$ and $\phi(n) \in \phi[N]$, $\phi(g)\phi(n)\phi(g)^{-1} = \phi(gng^{-1}) \in \phi[N]$. If $\phi(x) \in N'$ and $g \in G$, then $\phi(gxg^{-1}) = \phi(g)\phi(x)\phi(g)^{-1} \in N'$. ■

So normal subgroups of G/N correspond to normal subgroups of G containing N. A **maximal normal subgroup** of G is a proper normal subgroup contained in no larger proper normal subgroup.

Theorem 3.15 (15.18)

M is a maximal normal subgroup of G iff G/M is simple.

Proof. By Theorem 3.14 applied to $\gamma : G \to G/M$, a proper nontrivial normal subgroup of G/M has inverse image a normal subgroup strictly between M and G, and conversely a normal N with $M \subsetneq N \subsetneq G$ gives the proper nontrivial normal subgroup $\gamma[N]$ of G/M. ■

Theorem 3.16 (15.20)

The commutators $aba^{-1}b^{-1}$ generate a subgroup C of G (the **commutator subgroup**), C is normal, and for $N \trianglelefteq G$, G/N is abelian iff $C \leq N$.

Proof. C is a subgroup by definition (Theorem 1.18); the inverse of a commutator $aba^{-1}b^{-1}$ is the commutator $bab^{-1}a^{-1}$, so C consists of finite products of commutators. For $x \in C$ and $g \in G$, $gxg^{-1} = (gxg^{-1}x^{-1})x \in C$, so C is normal. G/N is abelian iff $abN = baN$ for all a, b iff $a^{-1}b^{-1}ab \in N$ for all a, b iff all commutators lie in N iff $C \leq N$. ■

3.4 Group action on a set (§16)

Motivation. Groups matter because they act: symmetries move points, permutations move letters, a group moves its own elements by conjugation. Formalizing actions gives the orbit–stabilizer theorem, a counting tool that will drive the Sylow theorems in Chapter 7.

An **action** of G on a set X is a map $G \times X \to X$, $(g, x) \mapsto gx$, with $ex = x$ and $(g_1g_2)x = g_1(g_2x)$; X is a **G-set**.

Theorem 3.17 (16.3)

For a G-set X and $g \in G$, $\sigma_g(x) = gx$ is a permutation of X, and $\phi : G \to S_X$, $\phi(g) = \sigma_g$, is a homomorphism with $\phi(g)(x) = gx$. Conversely, every homomorphism $G \to S_X$ defines an action by $gx = \phi(g)(x)$.

Proof. $\sigma_{g^{-1}}\sigma_g(x) = g^{-1}(gx) = ex = x$ and similarly $\sigma_g\sigma_{g^{-1}} = \mathrm{id}$, so σ_g is a bijection. $\sigma_{g_1g_2}(x) = (g_1g_2)x = g_1(g_2x) = \sigma_{g_1}\sigma_{g_2}(x)$. The converse is a direct check. ■

The action is **faithful** if only e acts as the identity, and **transitive** if for all x_1, x_2 some g has $gx_1 = x_2$. For $x \in X$, $G_x = \{g : gx = x\}$ is the **isotropy subgroup**, and $X_g = \{x : gx = x\}$.

Theorem 3.18 (16.12)

G_x is a subgroup of G for each $x \in X$.

Proof. $ex = x$; if $gx = hx = x$, then $(gh)x = g(hx) = x$ and $g^{-1}x = g^{-1}(gx) = x$. ■

Theorem 3.19 (16.14)

For $x_1, x_2 \in X$ let $x_1 \sim x_2$ iff $gx_1 = x_2$ for some $g \in G$. This is an equivalence relation on X.

Proof. $ex_1 = x_1$; if $gx_1 = x_2$ then $g^{-1}x_2 = x_1$; if $gx_1 = x_2$ and $hx_2 = x_3$ then $(hg)x_1 = x_3$. ■

The cells are the **orbits** $Gx = \{gx : g \in G\}$.

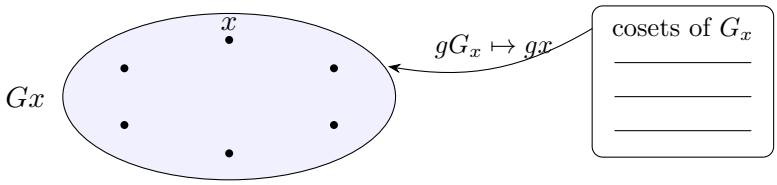


Figure. *Orbit–stabilizer: the points of the orbit Gx correspond one-to-one to the left cosets of the stabilizer G_x, so $|Gx| = (G : G_x)$.*

Theorem 3.20 (16.16, orbit–stabilizer)

$|Gx| = (G : G_x)$. If G is finite, $|Gx|$ divides $|G|$.

Proof. $gG_x \mapsto gx$ is well defined and one-to-one: $g_1x = g_2x$ iff $g_1^{-1}g_2 \in G_x$ iff $g_1G_x = g_2G_x$. It is onto Gx. Then use Lagrange. ■

Example 3.4

G acts on itself by left multiplication (transitive, faithful: Cayley) and by conjugation $g \cdot x = gxg^{-1}$, whose orbits are the conjugacy classes and whose isotropy groups are centralizers. S_n acts on $\{1, \ldots, n\}$; the rotation group of a cube acts on its 6 faces with isotropy groups of order 4, so it has order 24.

3.5 Applications of G-sets to counting (§17)

Motivation. How many essentially different ways are there to colour the faces of a cube? Burnside's formula answers such questions by counting orbits as the average number of fixed points.

Theorem 3.21 (17.1, Burnside's formula)

If G is finite, X a finite G-set and r the number of orbits, then $r \cdot |G| = \sum_{g \in G} |X_g|$.

Proof. Count $N = |\{(g, x) : gx = x\}|$ in two ways: $N = \sum_g |X_g| = \sum_x |G_x|$. By Theorem 3.20, $|G_x| = |G|/|Gx|$, so $\sum_{x \in \mathcal{O}} |G_x| = |G|$ for each orbit $\mathcal{O}$, and $N = r|G|$. ■

Corollary 3.22 (17.2)

The number of orbits is $\frac{1}{|G|} \sum_{g \in G} |X_g|$.

Example 3.5

Colouring the 6 faces of a cube with 6 different colours, up to rotation: only the identity fixes a colouring, so there are $6!/24 = 30$ distinguishable cubes. Necklaces of n beads with k colours up to rotation number $\frac{1}{n} \sum_{d|n} \phi(d) k^{n/d}$.

3.6 Selected exercises

Motivation. The first three exercises from the book are standard ways of proving that a subgroup is normal or that a group is abelian, using inner automorphisms, factor groups and Lagrange's theorem; the last is a Burnside count in which colours may repeat.

Exercise 3.1 (§14, Ex. 34; a unique subgroup of given order)

Show that if a finite group G has exactly one subgroup H of a given order, then $H \trianglelefteq G$.

Solution.

Step 1. For $g \in G$ the inner automorphism $i_g(x) = gxg^{-1}$ is a homomorphism, and it is a bijection with inverse $i_{g^{-1}}$.

Step 2. By Theorem 3.1(3), $i_g[H] = gHg^{-1}$ is a subgroup of G, and $|gHg^{-1}| = |H|$ since i_g is one-to-one.

Step 3. By uniqueness, $gHg^{-1} = H$ for every $g \in G$, so H is normal by Theorem 3.10. ■

Exercise 3.2 (§14, Ex. 30; powers in the factor group)

Let $H \trianglelefteq G$ with finite index $m = (G : H)$. Show that $a^m \in H$ for every $a \in G$.

Solution. The factor group G/H (Corollary 3.7) has exactly m elements. By Theorem 2.11 the order of aH divides m, so $(aH)^m$ is the identity H of G/H. Since coset multiplication gives $(aH)^m = a^m H$, we get $a^m H = H$, i.e. $a^m \in H$. ■

Exercise 3.3 (§15, Ex. 37–38; cyclic modulo the center)

The **center** of G is $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$.
(a) Show that $Z(G) \trianglelefteq G$, and that if $G/Z(G)$ is cyclic, then G is abelian.
(b) Let p, q be primes. Show that a nonabelian group G of order pq has trivial center.

Solution.

(a)

Step 1. $e \in Z(G)$. If $z, w \in Z(G)$, then $(zw)g = zgw = g(zw)$, and multiplying $zg = gz$ by z^{-1} on both sides gives $gz^{-1} = z^{-1}g$. So $Z(G) \leq G$ (Theorem 1.8), and $gzg^{-1} = z \in Z(G)$ shows it is normal (Theorem 3.10).

Step 2. Write $Z = Z(G)$ and suppose $G/Z = \langle gZ \rangle$. For $x, y \in G$ we have $xZ = g^i Z$ and $yZ = g^j Z$ for some $i, j \in \mathbb{Z}$, so $x = g^i z$ and $y = g^j w$ with $z, w \in Z$. Since z, w commute with everything,

$$xy = g^i z g^j w = g^{i+j} zw = g^j w g^i z = yx.$$

(b) By Lagrange, $|Z(G)| \in \{1, p, q, pq\}$. It is not pq, as G is nonabelian. If $|Z(G)|$ were p or q, then $|G/Z(G)|$ would be the prime q or p, so $G/Z(G)$ would be cyclic (Corollary 2.10) and G abelian by (a), a contradiction. (This also covers $p = q$.) Hence $Z(G) = \{e\}$. ■

Exercise 3.4 (§17, Ex. 5; colouring a cube with repetition)

Each face of a cube is painted with one of 8 colours, repetitions allowed. How many distinguishable cubes (up to rotation) are there?

Solution.

Step 1. The rotation group G of the cube ($|G| = 24$, Example 3.4) consists of: the identity; 6 rotations by $\pm 90°$ and 3 by $180°$ about axes through centres of opposite faces; 8 rotations by $\pm 120°$ about axes through opposite vertices; and 6 rotations by $180°$ about axes through midpoints of opposite edges.

Step 2. Let X be the set of 8^6 colourings. A colouring is fixed by g iff it is constant on each cycle of the permutation of the faces induced by g; so $|X_g| = 8^{c(g)}$, $c(g)$ the number of such cycles. The

cycle counts are 6 (identity), 3 ($\pm 90^\circ$ face: two fixed faces and a 4-cycle), 4 (180° face: two fixed faces and two 2-cycles), 2 (vertex: two 3-cycles) and 3 (edge: three 2-cycles).

Step 3. By Corollary 3.22, the number of orbits is

$$\frac{1}{24}\left(8^6 + 6 \cdot 8^3 + 3 \cdot 8^4 + 8 \cdot 8^2 + 6 \cdot 8^3\right) = \frac{281088}{24} = 11712.$$

■